Three Server Architecture – Deploying a Node.js Application Using Nginx

Server 1 – Build Server

Purpose

The Build Server is used to:

- Clone source code from Git
- Install dependencies
- Build the application
- Create ZIP artifact
- Upload artifact to Nexus Server

Step 1: Clone the Source Code

Install Git if not installed:

sudo apt update
sudo apt install git -y

Clone the repository:

git clone:

```
ubuntu@Build-server:~$ git clone https://github.com/Akiranred/AngularCalculator.git
Cloning into 'AngularCalculator'...
remote: Enumerating objects: 39, done.
remote: Total 39 (delta 0), reused 0 (delta 0), pack-reused 39 (from 1)
Receiving objects: 100% (39/39), 107.66 KiB | 8.28 MiB/s, done.
ubuntu@Build-server:~$
```

Step 2: Install Node.js and NPM

Install Node.js:

sudo apt install nodejs -y

Install NPM:

sudo apt install npm -y

Verify installation:

```
ubuntu@Build-server:~/AngularCalculator$ npm --version
9.2.0
ubuntu@Build-server:~/AngularCalculator$ nodejs --version
v22.22.1
ubuntu@Build-server:~/AngularCalculator$
```

Step 3: Install Project Dependencies

npm install

This installs all required packages from package.json.

Step 4: Build the Application

For Angular:

```
ubuntu@Build-server:~$ sudo npm install -g @angular/cli

added 288 packages in 10s

70 packages are looking for funding
  run `npm fund` for details
```

ng build

```
ubuntu@Build-server:~/AngularCalculator$ export NODE_OPTIONS=--openssl-legacy-provider
ubuntu@Build-server:~/AngularCalculator$ ng build
 10% building 4/4 modules 0 active(node:9717) [DEP0111] DeprecationWarning: Access to process.binding('http_parser') is
deprecated.
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:9717) [DEP0111] DeprecationWarning: Access to process.binding('http_parser') is deprecated.
(node:9717) [DEP0060] DeprecationWarning: The `util._extend` API is deprecated. Please use Object.assign() instead.

Date: 2026-05-18T13:27:18.546Z
Hash: df285638c9c085024033
Time: 11222ms
chunk {es2015-polyfills} es2015-polyfills.js, es2015-polyfills.js.map (es2015-polyfills) 285 kB [initial] [rendered]
chunk {main} main.js, main.js.map (main) 14.7 kB [initial] [rendered]
chunk {polyfills} polyfills.js, polyfills.js.map (polyfills) 236 kB [initial] [rendered]
chunk {runtime} runtime.js, runtime.js.map (runtime) 6.08 kB [entry] [rendered]
chunk {styles} styles.js, styles.js.map (styles) 973 kB [initial] [rendered]
chunk {vendor} vendor.js, vendor.js.map (vendor) 3.2 MB [initial] [rendered]
ubuntu@Build-server:~/AngularCalculator$
```

For React:

The build files are generated inside:

dist/

```
ubuntu@Build-server:~/AngularCalculator$ cd dist/
ubuntu@Build-server:~/AngularCalculator/dist$ ls
angularCalc  calculator-v1.zip
ubuntu@Build-server:~/AngularCalculator/dist$ |
```

Step 5: Create ZIP File

Compress the build folder:

zip -r calculator-v1.zip dist/

```
ubuntu@Build-server:~/AngularCalculator/dist$ zip -r calculator-v1.zip .
  adding: angularCalc/ (stored 0%)
  adding: angularCalc/polyfills.js (deflated 81%)
  adding: angularCalc/index.html (deflated 58%)
  adding: angularCalc/runtime.js.map (deflated 70%)
  adding: angularCalc/styles.js (deflated 81%)
  adding: angularCalc/styles.js.map (deflated 81%)
  adding: angularCalc/main.js.map (deflated 69%)
  adding: angularCalc/main.js (deflated 74%)
```

Step 6: Upload ZIP File to Nexus Server

Upload artifact using curl:

curl -v -u username:password --upload-file calculator-v1.zip
http://3.80.59.36:8081/repository/angular-builds/calculator-v1.zip

```
ubuntu@Build-server:~/AngularCalculator/dist$ curl -v -u admin:admin123 \--upload-file calculator-v1.zip \http://3.80.5
.36:8081/repository/angular-builds/calculator-v1.zip
*   Trying 3.80.59.36:8081...
* Established connection to 3.80.59.36 (3.80.59.36 port 8081) from 172.31.26.161 port 39114
* using HTTP/1.x
* Server auth using Basic with user 'admin'
> PUT /repository/angular-builds/calculator-v1.zip HTTP/1.1
> Host: 3.80.59.36:8081
> Authorization: Basic YWRtaW46YWRtaW4xMjM=
> User-Agent: curl/8.18.0
> Accept: */*
> Content-Length: 1808694
> Expect: 100-continue
>
< HTTP/1.1 100 Continue
<
* upload completely sent off: 1808694 bytes
< HTTP/1.1 201 Created
< Server: Nexus/3.85.0-03 (COMMUNITY)
< X-Content-Type-Options: nosniff
< Content-Security-Policy: default-src http: data: blob: 'unsafe-inline'; script-src http: 'unsafe-inline' 'unsafe-eval
< X-XSS-Protection: 1; mode=block
< Content-Length: 0
<
* Connection #0 to host 3.80.59.36:8081 left intact
ubuntu@Build-server:~/AngularCalculator/dist$ |
```

Server 2 – Nexus Repository Server

Purpose

The Nexus Server is used to:

- Store build artifacts
- Manage versions
- Provide download access to deployment server

Step 1: Install Java

sudo apt update
sudo apt install nodejs npm

```
ubuntu@Nexus-server:~$ sudo apt install nodejs npm
Installing:
  nodejs  npm

Installing dependencies:
  alacritty                                    node-graceful-fs
  build-essential                              node-growl
  bzip2                                        node-gyp
  cpp                                          node-has-flag
  cpp-15                                       node-has-unicode
  cpp-15-x86-64-linux-gnu                      node-has-value
  cpp-x86-64-linux-gnu                         node-has-values
  dpkg-dev                                     node-hosted-git-info
  eslint                                       node-http-proxy-agent
  fakeroot                                     node-https-proxy-agent
  g++                                          node-iconv-lite
  g++-15                                       node-icss-utils
  g++-15-x86-64-linux-gnu                      node-ieee754
  g++-x86-64-linux-gnu                         node-iferr
  gcc                                          node-ignore
  gcc-15                                       node-imurmurhash
  gcc-15-base                                  node-indent-string
  gcc-15-x86-64-linux-gnu                      node-inherits
  gcc-x86-64-linux-gnu                         node-ini
  gyp                                          node-interpret
  handlebars                                   node-ip
```

Step 2: Download Nexus Repository

Wget

```
Last login: Mon May 18 14:57:01 2026 from 104.28.157.147
ubuntu@Deploy-server:~$ wget http://admin:admin123@3.80.59.36:8081/repository/angular-builds/calculator-v1.zip
--2026-05-18 15:04:38--  http://admin:*password*@3.80.59.36:8081/repository/angular-builds/calculator-v1.zip
Connecting to 3.80.59.36:8081... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1808694 (1.7M) [application/zip]
Saving to: ‘calculator-v1.zip’

calculator-v1.zip            100%[=================================================>]   1.72M  --.-KB/s    in 0.01s

2026-05-18 15:04:38 (134 MB/s) - ‘calculator-v1.zip’ saved [1808694/1808694]
```

Step 3: Unzip Nexus

sudo unzip calculator-v1.zip

```
ubuntu@Deploy-server:~$ unzip calculator-v1.zip
Archive:  calculator-v1.zip
   creating: angularCalc/
  inflating: angularCalc/polyfills.js
  inflating: angularCalc/index.html
  inflating: angularCalc/runtime.js.map
  inflating: angularCalc/styles.js
  inflating: angularCalc/styles.js.map
  inflating: angularCalc/main.js.map
  inflating: angularCalc/main.js
  inflating: angularCalc/polyfills.js.map
  inflating: angularCalc/vendor.js
  inflating: angularCalc/es2015-polyfills.js
  inflating: angularCalc/runtime.js
  inflating: angularCalc/es2015-polyfills.js.map
  inflating: angularCalc/vendor.js.map
  inflating: angularCalc/favicon.ico
ubuntu@Deploy-server:~$ ls
```

Step 4: Start Nexus Server

Go to Nexus bin directory:

cd nexus/bin

Start Nexus:

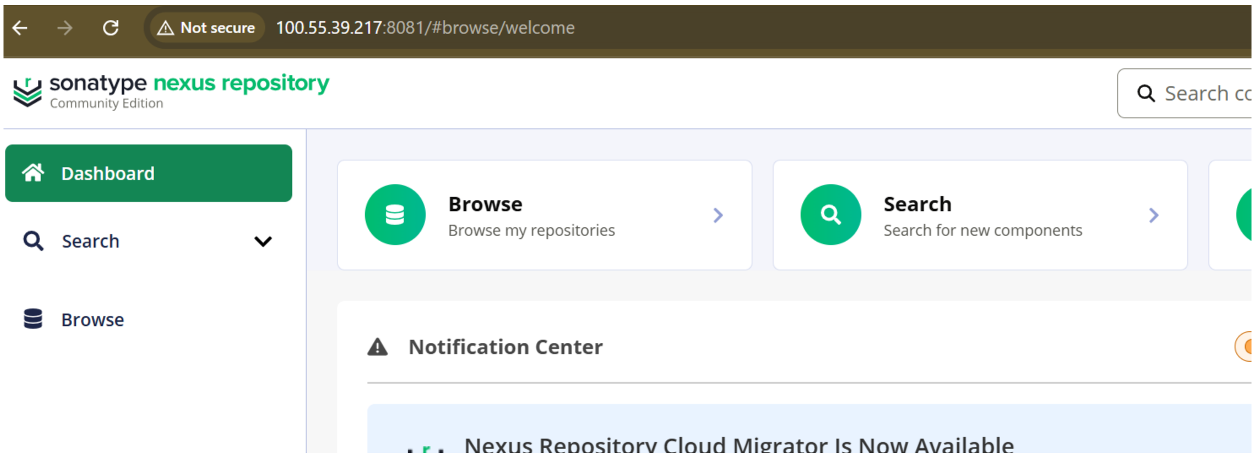
./nexus start

```
ubuntu@Nexus-server:~/nexus-3.85.0-03/bin$ ./nexus start
Starting nexus
ubuntu@Nexus-server:~/nexus-3.85.0-03/bin$
```

Step 5: Access Nexus UI

Open browser:

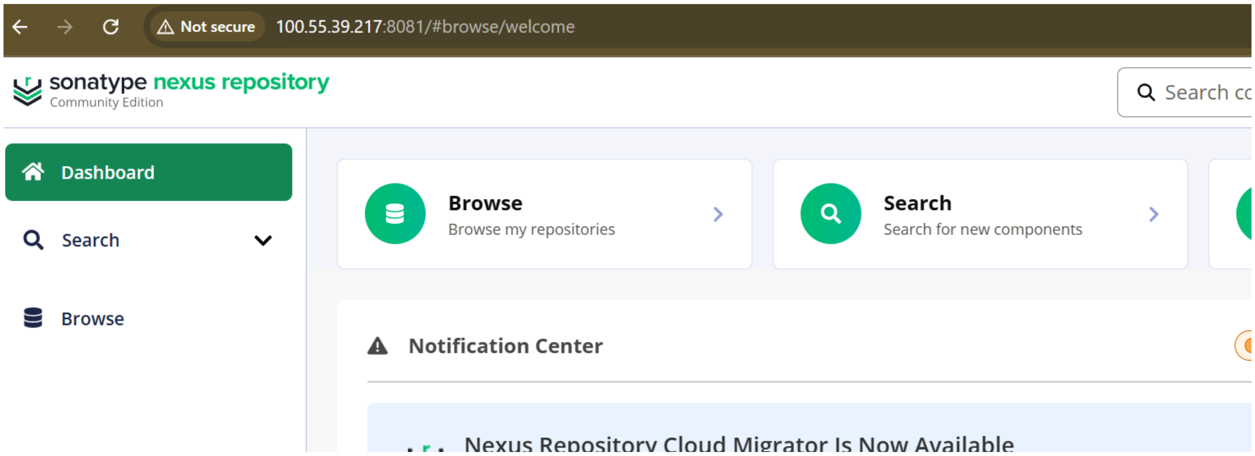
http://100.55.39.217:8081



Step 6: Create Repository

Inside Nexus UI:

1. Login to Nexus
2. Go to Settings
3. Click Repositories
4. Create Repository
5. Select Hosted Repository
6. Enter Repository Name
7. Save



Server 3 – Deployment Server

Purpose

The Deployment Server is used to:

- Download build artifact from Nexus
- Extract build files
- Deploy the application

Step 1: Install Nginix

sudo apt update
sudo apt install nginix

```
ubuntu@Deploy-server:~$ sudo apt install nginx
Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap  nginx-doc  ssl-cert

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 6
  Download size: 662 kB
  Space needed: 1874 kB / 3966 MB available
```

Step 2: Download ZIP File from Nexus

wget http://<nexus-server>:8081/repository/<repo-name>/application-build.zip

Example:

wget

```
103  wget http://admin:admin123@3.80.59.36:8081/repository/angular-builds/calculator-v1.zip
104  ls
```

Step 3: Extract ZIP File

unzip calculator-v1.zip

```
ubuntu@Deploy-server:~$ unzip calculator-v1.zip
```

Step 4: Deploy the Application

Copy files to deployment directory:

cp -r dist/* /var/www/html/

```
ubuntu@Deploy-server:~/angularCalc$ cd
ubuntu@Deploy-server:~$ sudo rm -rf /var/www/html/*
ubuntu@Deploy-server:~$ sudo cp -r angularCalc/* /var/www/html/
ubuntu@Deploy-server:~$ sudo systemctl restart nginx
ubuntu@Deploy-server:~$ ls
angularCalc  apache-tomcat-9.0.118  apache-tomcat-9.0.118.tar.gz  calculator-v1.zip  devops.pem
```

Step 5: Restart the nginx

```
ubuntu@Deploy-server:~/angularCalc$ cd
ubuntu@Deploy-server:~$ sudo rm -rf /var/www/html/*
ubuntu@Deploy-server:~$ sudo cp -r angularCalc/* /var/www/html/
ubuntu@Deploy-server:~$ sudo systemctl restart nginx
ubuntu@Deploy-server:~$ ls
angularCalc  apache-tomcat-9.0.118  apache-tomcat-9.0.118.tar.gz  calculator-v1.zip  devops.pem
```

Step 6: Final Output